TRACESMITH — SYSTEM ARCHITECTURE

GPU Profiling & Replay · C++17 core + pybind11 · source: docs/architecture.drawio · github.com/chenxingqiang/TraceSmith

APPLICATION LAYER

C++ Application
user binary

Python Application
import tracesmith

C++ CLI
cli/main.cpp

Python CLI
python/tracesmith/cli.py

External Tools
Perfetto UI / Chrome

PYTHON BINDINGS

pybind11 module
python/src/bindings.cpp

_tracesmith.so
native extension

python/tracesmith/__init__.py
package API

native

TRACESMITH CORE LIBRARY (C++17) include/tracesmith + src/

CUPTIProfiler
src/capture/cupti_profiler.cpp

MCPTIProfiler
src/capture/mcpti_profiler.cpp

MetalProfiler
src/capture/metal_profiler.mm

AscendProfiler
src/capture/ascend_profiler.cpp

ROCmProfiler
src/capture/rocm_profiler.cpp

BPFTracer
src/capture/bpf_tracer.cpp

MemoryProfiler
src/capture/memory_profiler.cpp

events

RingBuffer SPSC
common/ring_buffer.hpp

Types / TraceEvent
src/common/types.cpp

StackCapture
src/common/stack_capture.cpp

XRayImporter
src/common/xray_importer.cpp

SBTWriter/Reader
src/format/sbt_format.cpp

GPUStateMachine
src/state/gpu_state_machine.cpp

replay

InstructionStream
src/state/instruction_stream.cpp

TimelineBuilder
src/state/timeline_builder.cpp

TimelineViewer
src/state/timeline_viewer.cpp

PerfettoExporter
src/state/perfetto_exporter.cpp

PerfettoProto
src/state/perfetto_proto_exporter.cpp

ReplayEngine
src/replay/replay_engine.cpp

StreamScheduler
src/replay/stream_scheduler.cpp

OperationExecutor
src/replay/operation_executor.cpp

DeterminismChecker
src/replay/determinism_checker.cpp

FrameCapture
src/replay/frame_capture.cpp

GPUTopology
src/cluster/gpu_topology.cpp

MultiGPUProfiler
src/cluster/multi_gpu_profiler.cpp

TimeSync NTP/PTP
src/cluster/time_sync.cpp

NCCLTracker
src/cluster/nccl_tracker.cpp

GDB RSP
src/gdb/rsp_handler.cpp

TracyExporter
src/tracy/tracy_exporter.cpp

GPU BACKENDS & SYSTEM (createProfiler + nsys / mcTracer / xctrace / msprof)

NVIDIA CUDA
CUPTI SDK + nsys

Apple Metal
Metal API + xctrace

MetaX MACA
MCPTI + mcTracer

Huawei Ascend
CANN + msprof

AMD ROCm
roctracer

Linux eBPF
kernel events

libunwind
host stacks

Perfetto SDK
protobuf

OUTPUT FORMATS

trace.sbt
SBT binary

trace.json
Perfetto JSON

trace.perfetto
protobuf

graph.dot
Graphviz

ASCII timeline
terminal

comm matrix
NCCL heatmap

v0.10+ · capture→ring→SBT→state/timeline→replay/export · files from include/tracesmith and src/